

1 Question

The Harris-Benedict equation estimates the number of calories your body needs to maintain your weight if you do no exercise. This is called your basal metabolic rate or BMR.

The calories needed for a woman to maintain her weight is:

$$BMR = 655.1 + (9.6 \times w) + (1.85 \times h) - (4.7 \times a)$$

The calories needed for a man to maintain his weight is:

$$BMR = 66.5 + (13.75 \times w) + (5 \times h) - (6.75 \times a)$$

where w is the weight of the person in kg, h is the height of the person in cm, and a is the age of the person.

A typical chocolate bar will contain around 230 calories. Write a program that allows the user to input their weight, height, and age. The program should then output the number of chocolate bars that should be consumed to maintain one's weight for both a woman and a man.

```
What is your age?
```

```
25
```

```
What is your height?
```

```
185
```

```
What is your weight?
```

```
87
```

```
If you're female, you should eat 7.45 candy
```

```
If you're male, you should eat 8.78 candy
```

2 Question

A palindrome is a sequence of characters that reads the same backward as forward. For example, each of the following four-digit integers is a palindrome: 1221, 5555, 4554 and 8118. Write a program that reads in a four-digit integer and determines whether it is a palindrome. If the number is not four digits long, display an error message and exit the program. Otherwise, check if it's a palindrome or not.

```
Enter number
```

```
1441
```

```
The number is palindrome
```

3 Question

Write a program which takes two integers a and b from the user. The program will print the number of integers which are divisible by 3 between $[a, b]$. Loops are not allowed in this problem.

a: 5

b: 126

Numbers divisible by 3: 41

4 Question

Write a program which takes a positive integer less than 16;

- prints "Divisible by 2", if the number is divisible by 2 (2, 4, 6, 8, 10, 12, 14).
- prints "Divisible by 3", if the number is divisible by 3 (3, 6, 9, 12, 15).
- prints "Prime", if the number is prime (2, 3, 5, 7, 11, 13)
- prints "Fibonacci", if the number is one of the fibonacci numbers (1, 2, 3, 5, 8, 13).

If one number satisfies more than one condition, all messages must be displayed. You must use switch-case construct in this problem.

```
Enter number: 2
Divisible by 2
Prime
Fibonacci
```

```
Enter number: 15
Divisible by 3
```

5 Question

Write a program which takes three integers (x, y, z) from the user and prints the harmonic mean of those three integers. The harmonic mean of three numbers is calculated as

$$H = \frac{1}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$$

Enter three numbers:

x: 2

y: 3

z: 6

Harmonic Mean: 1

Enter three numbers:

x: 4

y: 4

z: 8

Harmonic Mean: 1.6

6 Question

There are four dogs and three cats in our house. Not every dog likes every cat. First dog "Karabaş" likes first cat "Zeytin" and third cat "Pamuk". Second dog "Ares" likes second cat "Duman". Third dog "Max" likes all cats. Fourth dog "Thor" likes "Zeytin" and "Duman". Write a program which takes the index of the dog and index of the cat and displays "Yes" on the screen if the given dog likes the given cat, "No" otherwise". You must use switch-case construct in this problem.

```
Enter dog index: 1
Enter cat index: 3
Yes
```

```
Enter dog index: 4
Enter cat index: 3
No
```